

TOPIC NAME: Transformer Math

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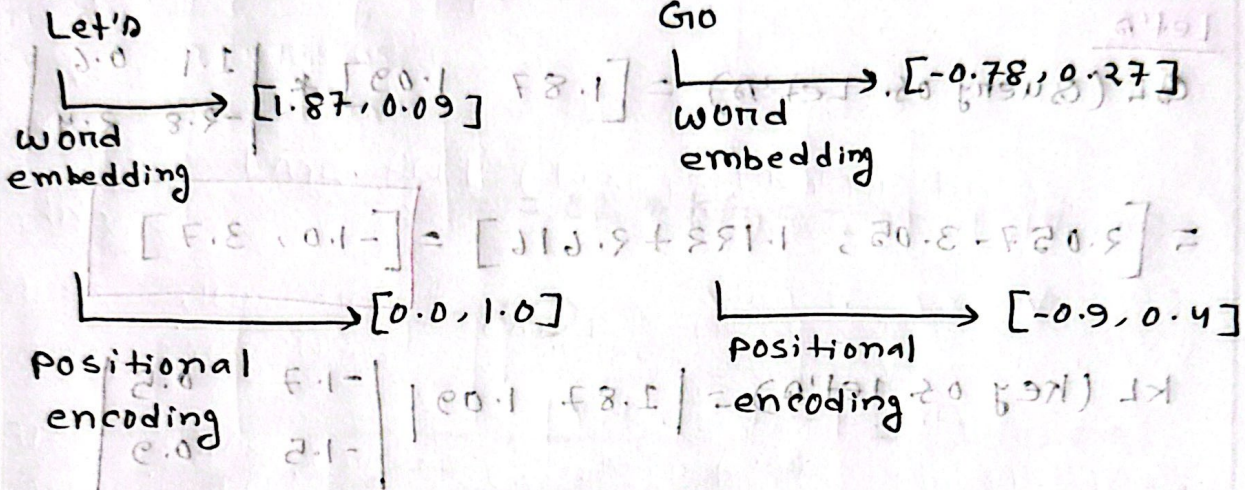
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* Input

Let's Go Translate to Vamos

Encoder part (For input)



Word embedding + positional encoding

$$\begin{bmatrix} 1.87 & 0.09 \\ 0.0 & 1.0 \end{bmatrix} + \begin{bmatrix} -0.78 & 0.27 \\ -0.9 & 0.4 \end{bmatrix} = \begin{bmatrix} 1.09 & 0.36 \\ -0.9 & 1.4 \end{bmatrix}$$

Let's Go

Now we will calculate attention

$$\begin{bmatrix} 2.0 & 1.1 \\ 0.0 & 2.1 \end{bmatrix} \begin{bmatrix} 1.09 & 0.36 \\ -0.9 & 1.4 \end{bmatrix} = \begin{bmatrix} 2.18 & 0.78 \\ -0.9 & 2.94 \end{bmatrix}$$

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Given weights for Query, key, Value:

$$Q = \begin{bmatrix} 1.1 & 0.6 \\ -2.8 & 2.4 \end{bmatrix}$$

$$K = \begin{bmatrix} -1.7 & 0.5 \\ -1.5 & 0.9 \end{bmatrix}$$

$$V = \begin{bmatrix} 1.5 & -1.0 \\ -0.3 & -0.2 \end{bmatrix}$$

Let's

$$Q_L (\text{Query of Let's}) = [1.87 \quad 1.09] \times \begin{bmatrix} 1.1 & 0.6 \\ -2.8 & 2.4 \end{bmatrix}$$

$$= [2.057 - 3.05, 1.122 + 2.616] = [-1.0, 3.7]$$

$$K_L (\text{key of Let's}) = [1.87 \quad 1.09] \times \begin{bmatrix} -1.7 & 0.5 \\ -1.5 & 0.9 \end{bmatrix}$$

$$= [-4.7, 1.9]$$

$$V_L (\text{value of Let's}) = [1.87 \quad 1.09] \times \begin{bmatrix} 1.5 & -1.0 \\ -0.3 & -0.2 \end{bmatrix}$$

$$= [2.5, -2.1]$$

Go

$$Q_G (\text{Query of Go}) = [-1.68 \quad 0.67] \times \begin{bmatrix} 1.1 & 0.6 \\ -2.8 & 2.4 \end{bmatrix}$$

$$= [-3.7, 0.6]$$

$$K_G (\text{key of Go}) = [-1.68 \quad 0.67] \times \begin{bmatrix} -1.7 & 0.5 \\ -1.5 & 0.9 \end{bmatrix}$$

$$= [1.9, -0.2]$$

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$$V_G (\text{value of } G_0) = \begin{bmatrix} -1.68 & 0.67 \end{bmatrix} \begin{bmatrix} 1.5 & -1.0 \\ -0.3 & -0.2 \end{bmatrix}$$

$$= \begin{bmatrix} -2.7 & 1.5 \end{bmatrix}$$

Attention for Let's:

$$\text{Let's with Let's} = 8L * KL = 11.7$$

$$\text{Let's with } G_0 = 8L * KG = -2.6$$

$$\text{Softmax} (11.7, -2.6)$$

$$\Rightarrow \frac{e^{11.7}}{e^{11.7} + e^{-2.6}}, \frac{e^{-2.6}}{e^{11.7} + e^{-2.6}} = \begin{bmatrix} 1.0 & 0.0 \end{bmatrix}$$

↳ contribution of G_0
↳ contribution of Let's

$$\text{Now } VL \times 1 = \begin{bmatrix} 2.5 & -2.1 \end{bmatrix}$$

$$VG \times 0 = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

$$+ \begin{bmatrix} 2.5 & -2.1 \end{bmatrix}$$

$$+ \begin{bmatrix} 1.87 & 1.09 \end{bmatrix}$$

$$\begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix} \begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix} = \begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix}$$

↳ Attention for Let's

$$\begin{bmatrix} 0.1 & 0.1 \end{bmatrix} = \begin{bmatrix} 1.0 & 1.0 \\ 1.0 & 1.0 \end{bmatrix} \begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix} = \begin{bmatrix} 1.0 & 1.0 \end{bmatrix}$$

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Attention $\rightarrow$ $G_0: 1.0, 82.1 \rightarrow (0.0, 2.0, 2.0) \rightarrow 10V$

$$Q_G * K_G = -7.2$$

$$Q_G * K_L = 18.5 \quad F.8 - 1 =$$

$$\text{softmax}(18.5, -7.2) = [1.0, 0.0]$$

$\downarrow$ contribution of G_0
 $\rightarrow$ contribution of L_0

$$\text{So, } v_L * 1 = 2.5, -2.1$$

$$v_G * 0 = 0.0$$

$$(2.5 - 1.1) \times \text{softmax}$$

$$\begin{aligned} & \frac{[0.0, 0.0] + [2.5, -2.1]}{2.5 - 1.1} \leftarrow F.11 \\ & + \frac{[-1.68, 0.67]}{2.5 - 1.1} \leftarrow F.11 \\ & \{0.82, -1.43\} \quad \checkmark \end{aligned}$$

$\rightarrow$ Attention $\rightarrow$ G_0

$$[0, 0] = 0 \times 10V$$

Decoden Part

$$\text{EOS } [2.70, -1.34] + \text{Positional} \rightarrow \text{encoding } [0.0, 1.0]$$

$$[2.70, -0.34]$$

$$Q = \begin{bmatrix} 0.4 & 0.4 \\ -0.3 & 0.1 \end{bmatrix} \quad K = \begin{bmatrix} 0.4 & -0.7 \\ -0.4 & 0.3 \end{bmatrix} \quad V = \begin{bmatrix} -1.1 & -0.7 \\ -0.4 & 1.3 \end{bmatrix}$$

$$Q E = [2.70 \quad -0.34] \begin{bmatrix} 0.4 & 0.4 \\ -0.3 & 0.1 \end{bmatrix} = [1.2, 1.0]$$

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$$K E = \begin{bmatrix} 1.5 & -0.3 \\ 0.3 & -1.0 \end{bmatrix} \begin{bmatrix} 1.1 & -0.7 \\ 0.4 & 0.3 \end{bmatrix} = \begin{bmatrix} 1.2 & 1.8 \\ -2.8 & -2.3 \end{bmatrix}$$

$$V E = \begin{bmatrix} 1.5 & -0.3 \\ 0.3 & -1.0 \end{bmatrix} \begin{bmatrix} -1.1 & -0.7 \\ -0.4 & 0.3 \end{bmatrix} = \begin{bmatrix} -2.8 & -2.3 \end{bmatrix}$$

$$S E \times K E = -0.4$$

$$\text{softmax}(-0.4) = 1.0$$

$$E O S [2.70, -0.34] + V E \times 1.0 = [-0.1, -2.6]$$

Decoder Encoder cross attention calculation (EOS, let's, go)

Given,

$$Q = \begin{bmatrix} 1.5 & -0.3 \\ 0.3 & -1.0 \end{bmatrix} \begin{bmatrix} 1.1 & -0.7 \\ 0.4 & 0.3 \end{bmatrix} = \begin{bmatrix} 1.2 & 1.8 \\ -2.8 & -2.3 \end{bmatrix}$$

What we got previously:

$$E O S [-0.1, -2.6] + [2.9, -1.0] = [2.8, -3.6]$$

$$\text{Let's} = [4.3, -1.01]$$

$$G O = [0.82, -1.43]$$

$$S E = E O S \times Q = \begin{bmatrix} -0.1 & -2.6 \\ 1.5 & -0.3 \\ 0.3 & -1.0 \end{bmatrix} = \begin{bmatrix} -0.9 & 2.6 \end{bmatrix}$$

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$$kL = \begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix} \times \begin{bmatrix} 4.3 & -1.01 \\ -0.3 & -0.8 \end{bmatrix} = \begin{bmatrix} 4.3 & -1.01 \\ -0.3 & -0.8 \end{bmatrix}$$

$$vL = \begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix} \times \begin{bmatrix} 4.3 & -1.01 \\ -0.3 & -0.8 \end{bmatrix} = \begin{bmatrix} 4.3 & -1.01 \\ -0.3 & -0.8 \end{bmatrix}$$

$$kG = g_0 \times k = \begin{bmatrix} 0.82 & -1.43 \\ -0.3 & -0.8 \end{bmatrix} \times \begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix} = \begin{bmatrix} 0.82 & -1.43 \\ -0.3 & -0.8 \end{bmatrix}$$

$$vG = g_0 \times v = \begin{bmatrix} 0.82 & -1.43 \\ -0.3 & -0.8 \end{bmatrix} \times \begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix} = \begin{bmatrix} 0.82 & -1.43 \\ -0.3 & -0.8 \end{bmatrix}$$

$$gE \times kL = 9.5$$

$$gE \times kG = 4.1$$

$$\text{softmax}(9.5, 4.1) = \begin{bmatrix} 1.0 & 0.0 \end{bmatrix}$$

$$vL \times 1.0 + vG \times 0 = \begin{bmatrix} 4.3 & -1.01 \end{bmatrix}$$

$$EOS[-0.1, -2.6] + \begin{bmatrix} 4.3 & -1.01 \end{bmatrix} = \begin{bmatrix} 5.9 & -0.5 \end{bmatrix}$$

When we pass these values through Neural Network,

we got max value of vamos.

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Attention calculation in decoder (Vamos, EOS)

$$\text{Vamos} [-0.86, 2.37] + \text{positional encoding} [-0.9, 0.4] = [-0.86, 2.37]$$

$$Q = \begin{bmatrix} 0.4 & 0.4 \\ -0.3 & 0.1 \end{bmatrix} \quad K = \begin{bmatrix} 0.4 & -0.7 \\ -0.4 & 0.3 \end{bmatrix} \quad V = \begin{bmatrix} -1.1 & -0.7 \\ -0.4 & 1.3 \end{bmatrix}$$

$$\text{EOS} = [2.7, -0.34] \rightarrow \text{value after positional encoding}$$

$$KE = \text{EOS} \times K = [2.7, -0.34] \times \begin{bmatrix} 0.4 & -0.7 \\ -0.4 & 0.3 \end{bmatrix} = [1.2, -1.8]$$

$$VE = \text{EOS} \times V = [2.7, -0.34] \times \begin{bmatrix} -1.1 & -0.7 \\ -0.4 & 1.3 \end{bmatrix} = [-2.8, -2.3]$$

$$QV = \text{Vamos} \times Q = [-0.86, 2.37] \times \begin{bmatrix} 0.4 & 0.4 \\ -0.3 & 0.1 \end{bmatrix} = [-1.3, -0.1]$$

$$VV = \text{Vamos} \times V = [-0.86, 2.37] \times \begin{bmatrix} -1.1 & -0.7 \\ -0.4 & 1.3 \end{bmatrix} = [0.0, 3.7]$$

$$QV \times KE = -1.1 \quad QV \times KV = 1.4$$

$$\text{softmax}(-1.1, -1.4) = 0.1, 0.9$$

$$\text{Vamos} [-0.86, 2.37] + VE \times 0.1 + VV \times 0.9 = [-1.2, 5.5]$$

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Decoder Encoder cross attention (Vamos, Lets, go)

$$Q = \begin{bmatrix} 1.5 & -0.3 \\ 0.3 & -1.0 \end{bmatrix} \quad K = \begin{bmatrix} -1.1 & 0.3 \\ -0.3 & -0.8 \end{bmatrix} \quad V = \begin{bmatrix} 1.1 & 0.6 \\ -1.2 & -0.5 \end{bmatrix}$$

$$L = \begin{bmatrix} 1.1 & 0.6 \\ 1.0 & 0.0 \end{bmatrix} \quad G = \begin{bmatrix} 0.82 & 1.43 \end{bmatrix}$$

$$QV = \text{Vamos} \times Q = \begin{bmatrix} 1.2 & 5.5 \end{bmatrix} \quad \begin{bmatrix} 1.5 & -0.3 \\ 0.3 & -1.0 \end{bmatrix} = [-0.1, -5.1]$$

$$KL = \text{Lets} \times K = [-4.5, 2.1]$$

$$KG = \text{go} \times K = [-0.5, 1.4]$$

$$VL = \text{Lets} \times V = [6.0, 3.1]$$

$$VG = \text{go} \times V = [2.6, 1.2]$$

$$QV \times KL = -10.3 \quad QV \times KG = -7.1$$

$$\text{So } \max(-10.3, -7.1) = 0.0, 1.0$$

$$VL \times 0.0 + VG \times 1.0 = [2.6, 1.2]$$

$$\text{Vamos} [-1.2, 5.5] + [2.6, 1.2] = [1.4, 6.7]$$

$$e.0, 1.0 = (1.1 - 1.1) \times \text{Vamos}$$

$$[2.2, 5.1] = e.0 \times V + 1.0 \times G + [5.5, 0.8] \times \text{Vamos}$$